隐藏翻译（T）

Could we have babies in space?

【BBC精选：我们能在太空中生宝宝吗？】

Summary: NASA plans to send humans to Mars within decades, aiming for eventual independence from Earth. Space radiation poses significant risks to human reproduction, damaging sperm and egg cells. Animal studies show some promise, with mouse sperm producing healthy offspring despite DNA changes. Microgravity could cause developmental issues, affecting balance, bone density, and even surgery procedures. Children born in space may never adapt to Earth's gravity, raising ethical questions about consent and human identity. Solving these challenges is crucial for deep space exploration.

摘要： NASA计划在未来几十年内将人类送上火星，最终实现不依赖地球的独立生存。太空辐射对人类繁殖构成重大风险，会损害精子和卵细胞。动物研究显示了一些希望：小鼠精子虽出现DNA变化，仍能孕育健康后代。微重力环境可能导致发育问题，影响平衡感、骨密度甚至手术操作。在太空出生的孩子可能永远无法适应地球重力，这引发了关于人类身份和代际同意的伦理争议。解决这些挑战对于深空探索至关重要。

⏱ Estimated Reading Time: 7 min.

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Humans will travel to Mars within the next couple of decades.

人类将在未来几十年内前往#火星#。

That's according to NASA anyway.

至少美国宇航局是这样说的。

The ultimate goal to live entirely independent of Earth.

最终目标是完全独立于地球生活。

Scientists are already developing technology that will enable us to travel the incredibly long distances and keep us alive when we get there.

科学家们已经在开发技术，使我们能够完成这段极其漫长的旅程，并在到达时保持存活。

But if we really want to inhabit other worlds, a major issue needs to be addressed.

但如果我们真的想#居住#在其他星球上，就必须解决一个重大问题。

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How do we reproduce in space?

我们该如何在太空中繁殖?

If humans had a base on Mars, we couldn't keep popping back 225 million kilometres,

如果人类在#火星#上建立了基地，就不可能往返2.25亿公里，

a journey likely to take around seven months to have children on Earth.

花七个月的时间回地球生孩子。

Humans will need to have babies in space.

人类将需要在太空中生育。

But space is a hostile environment for the human body, with high levels of radiation.

但太空是对人体极其#敌对#的环境，充满了高水平的#辐射#。

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Something both sperm and eggshells are highly sensitive to.

而精子和卵细胞对辐射极其敏感。

NASA recently sent the first samples of frozen human sperm to space.

NASA最近将首批冷冻人类精子样本送入太空。

The frozen cells were successfully reactivated, but showed DNA damage and moved differently,

这些冷冻细胞成功被重新激活，但表现出DNA损伤，且运动方式不同，

which would likely reduce their chances of fertilizing an egg.

这可能会降低它们与卵子结合的几率。

More encouragingly, freeze-dried mouse sperm that spent several months in space, produced a root of healthy pops when fertilized back on Earth.

更令人鼓舞的是，经过数月太空旅程的冻干小鼠精子，在地球上受精后仍能诞下健康幼崽。

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Although the sperm DNA had changed, the damage was repaired upon contact with the female egg.

尽管精子DNA发生了变化，但在与卵子结合时损伤得到了修复。

For a fetus to develop in space, however, it would need to be protected from this harmful radiation.

然而，要让#胎儿#在太空中发育，就必须保护其免受这种有害的#辐射#。

Its rapid growth rate would make a baby particularly vulnerable to genetic mutations,

婴儿快速的生长速度会使其特别#易受# #基因#突变影响,

increasing the risk of childhood cancers.

从而增加儿童癌症的风险。

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And perhaps a significant as radiation in the long term is time spent away from Earth's gravity.

而长期来看，和#辐射#同样重要的问题是远离地球#重力#。

We know that low gravity causes astronauts' muscles to atrophy, bones to become less dens,

我们知道低#重力#会导致宇航员的肌肉#萎缩#、骨密度下降,

and the quantity of circulating blood in the body to decrease.

以及体内血液循环量的减少。

The first baby born off-planet Earth will most likely be born on Mars,

地球之外的第一个婴儿很可能会诞生在#火星#,

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where gravity is about 38% of Earth's, which is bound to affect both birth and development.

那里#重力#只有地球的约38%，必然会影响分娩和发育。

But if a child was born in microgravity,

但如果婴儿出生在微重力环境下,

for example on a space station, the impacts would be much more pronounced.

例如在空间站上，其影响将更加#明显#。

For the sake of argument, let's imagine this happening.

为了#讨论#起见，让我们设想一下这种情况。

It's thought the gravity helps shape parts of the inner ear that play an important role in balance and orientation.

人们认为#重力#有助于塑造内耳的一部分，而这部分在平衡和#方向感#中起着重要作用。

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Rat pops born after developing in space, couldn't sense which way was up or down,

在太空中发育的小#鼠#幼崽，无法分辨上下方向,

and microgravity would continue to pose challenges for childbirth itself.

微重力甚至会继续给分娩本身带来挑战。

On Earth, there is always the option of a C-section,

在地球上，我们总可以选择剖腹产,

but although medics have mended rat's tails and performed keyhole surgery on pigs, no one has ever been operated on in space.

但尽管医生修复过小#鼠#的尾巴，还曾对猪进行过微创手术，却从未有人在太空中接受过手术。

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Blood can spatter even more than it usually does on Earth, floating around unconstrained by gravity,

血液可能比地球上更容易#飞溅#，在失去#重力#约束的环境中四处漂浮，

or a cumpul into a kind of dome around a wound, making it hard to see what's going on.

或聚成一个#圆顶#状包裹在#伤口#周围，使医生难以观察。

Today, if an astronaut needs emergency surgery, they have to ditch their mission and return to Earth.

如今，如果一名#宇航员#需要紧急手术，他必须#放弃#任务并返回地球。

But presuming the birth was successful, a baby growing up in microgravity would look, move,

但假设分娩成功，在微重力中长大的婴儿在外貌、动作上，

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and behave very differently to one born on Earth.

以及行为上，都会与地球出生的婴儿截然不同。

There, a child would not learn to crawl, but would instead learn to float,

在那里，孩子不会学会#爬行#，而会学会漂浮，

propelling themselves using their arms.

并用手臂推动自己前进。

Without a strict exercise regime, it's possible their bones and muscles would be smaller in their lower body,

如果没有严格的锻炼#制度#，他们下半身的骨骼和肌肉可能会更小，

yet bigger and stronger in their upper body.

而上半身则可能更大更强壮。

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Like astronauts, fluids in their body could travel upwards into their chest and head,

就像宇航员一样，他们体内的液体可能会上移到胸部和头部，

giving them a puffy face.

让他们看起来脸部浮肿。

As a result, a child born and raised in space might never be able to live on Earth.

因此，一个在太空出生并长大的孩子可能永远无法在地球上生活。

It's possible they might not be able to walk, stand, or even breathe.

他们可能无法行走、站立，甚至无法呼吸。

Earth's gravity is so important to humans that it's been described as the identity of mankind.

地球的#重力#对人类如此重要，以至于被称为#人类#的身份标志。

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Without it, would a human subspecies emerge, only able to survive in space?

没有它，会不会出现只能在太空生存的人类#亚种#?

Assuming all the physical challenges can be overcome, there are other issues that would need to be considered.

即使所有生理难题都能克服，仍然有其他问题需要考虑。

What would be the impacts of raising a child in such an isolated and extreme environment?

在这样一个#孤立#而极端的环境中养育孩子，会带来怎样的影响?

While a mother may be able to consent, the child cannot.

母亲也许能给予#同意#，但孩子却不能。

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Is it ethical to raise human beings who may never be able to set foot on planet Earth?

抚养一个可能永远无法踏上地球的人类，这样做是否#符合伦理#?

Right now, having a baby in space may feel like something from a sci-fi film.

现在，在太空中生育听起来更像是科幻电影里的情节。

But ultimately, it's a puzzle we have to solve if we want to travel deeper into our solar system.

但归根结底，如果我们想深入探索太阳系，这就是一个必须解决的#难题#。